04/12/23 MS

USN. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]

[20EC71]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER - 2023

INDUSTRIAL MANAGEMENT AND OPERATION RESEARCH

Time: 3 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.
- **iii.** Use the graph sheets wherever necessary.

1. a. Write the mathematical model of linear programming problem. What are the assumptions of linear programming model? **07**

b. Use the graphical method to solve the following LP problem. **07**

Maximize $Z = 15x_1 + 10x_2$

subject to the constraints

(i) $4x_1 + 6x_2 \leq 360$ (ii) $3x_1 + 0x_2 \leq 180$ (iii) $0x_1 + 5x_2 \leq 200$ and $x_1, x_2 \geq 0$.

2. a. Apply MODI method to obtain optimal solution of transportation problem using the below data for minimization of total cost using VAM approach. **08**

	D_1	D_2	D_3	D_4	Supply
S_1	19	30	50	10	7
S_2	70	30	40	60	9
S_3	40	8	70	20	18
Demand	5	8	7	14	34

b. An automobile dealer wishes to put four repairmen to four different jobs. The repairmen have somewhat different kinds of skills and they exhibit different levels of efficiency from one job to another. The dealer has estimated the number of man-hours that would be required for each job-man combination. This is given in matrix form in the following table **06**

		Jobs			
		A	B	C	D
Men	1	5	3	2	8
	2	7	9	2	6
	3	6	4	5	7
	4	5	7	7	8

Find the optimal assignment that will result in minimum man-hours needed.

3. a. Products 1, 2, 3, 4 and 5 are to be processed on a machine. The set-up costs in rupees per change depend upon the product processed on the machine and the set-up to be made. These are given by the following data: **07**

$C12 = 16$, $C13 = 4$, $C14 = 12$, $C23 = 6$, $C24 = 5$, $C25 = 8$,

$C35 = 6$, $C45 = 20$; $Cij = Cji$, $Cij = \infty$ for $i = j$ for all values of i and j not given in the data. Find the optimum sequence of products in order to minimize the total set-up cost.

b. Define the following in sequencing problems, **07**

- **i.** Idle time on a machine
- **ii.** Total elapsed time
- **iii.** No passing rule

Also, state the assumptions made in the sequencing problems.

OR

4. a. A manufacturing company processes 6 different jobs on two machines A and B. Number of units of each job and its processing times on A and B are given below. Find the optimal sequence, the total minimum elapsed time and idle time for either machine. **07**

Job Number	*Number of Units of Each Job*	*Processing Time (in Minutes)*	
		Machine A	*Machine B*
1	3	5	8
2	4	16	7
3	2	6	11
4	5	3	5
5	2	9	7.5
6	3	6	14

b. Using the graphical method, calculate the minimum time needed to process jobs 1 and 2 on five machines A, B, C and D, i.e. for each machine find the job that should be done first. Also, calculate the total time needed to complete both jobs. **07**

			Machines			
Job 1	*Sequence*	:	*A*	*B*	*C*	*D*
	Time (hrs)	:	6	8	4	12
Job 2	*Sequence*	:	*B*	*C*	*A*	*D*
	Time (hrs)	:	10	8	6	4

5. a. What is pay-off matrix in game theory? State the assumptions of the game. **06**

b. Solve the game whose payoff matrix is given below: **08**

	Player *B*			
Player *A*	B_1	B_2	B_3	B_4
A_1	3	2	4	0
A_2	3	4	2	4
A_3	4	2	4	0
A_4	0	4	0	8

OR

6. a. Find the range of values of p and q that will render the entry (2, 2) a saddle point for the game **05**

Player B

Player A	B_1	B_2	B_3
A_1	2	4	5
A_2	10	7	q
A_3	4	p	6

b. Two firms A and B make colour and black & white television sets. Firm A can make either 150 colour sets in a week or an equal number of black & white sets, and make a profit of Rs 400 per colour set, or 150 colour and 150 black & white sets, or 300 black & white sets per week. It also has the same profit margin on the two sets as A. Each week there is a market of 150 colour sets and 300 black & white sets and the manufacturers would share market in the proportion in which they manufacture a particular type of set. **09**

Write the pay-off matrix of A per week. Obtain graphically A's and B's optimum strategies and value of the game.

7. a. Define management. Briefly explain the management functions. **08**

b. What is the difference between management and administration? How far is this distinction justified in your opinion? **06**

OR

8. a. What are dysfunctional effects of informal organization? How to minimize these effects. **07**

b. What is recruitment? Explain the sources of recruitment? **07**